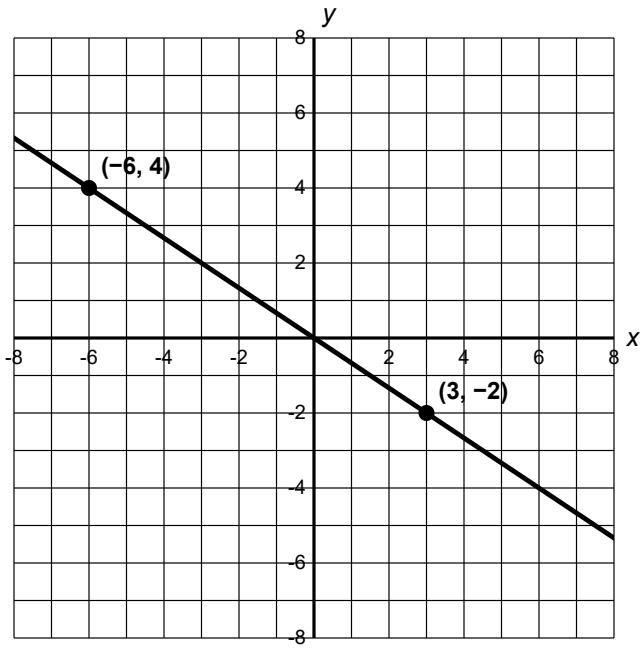
Topic 2 Review Test · ANSWER KEY

Algebra I · Module 2, Topic 2: Transforming and Comparing Linear Functions · Monday, Nov 2

Directions: This review has the same kinds of questions as the Topic 2 test. Read each question carefully. Show your work. **Select TWO** means choose exactly two answers. Finish Part 1 first; Part 2 is extra practice.

Part 1 · Just like the test (Questions 1–10)

1. The graph of linear function *m* is shown on the grid.



Given $f(x) = x$ and $m(x) = af(x)$, what is the value of a?

$a =$ $-\frac{2}{3}$

slope = (−2 − 4) ÷ (3 − (−6)) = −6/9 = −2/3, and a is the slope of m(x) = a·x

2. The cost of renting a bounce house, *y*, for *x* hours at a birthday party can be modeled using a linear function. The rental company charges a fixed fee of $75, plus an additional $20 per hour. The bounce house can be rented for a maximum of 5 hours per day. For one day of rental, what is the range of the function for this situation?

A. $0 \le x \le 5$

B. $20 \le y \le 175$

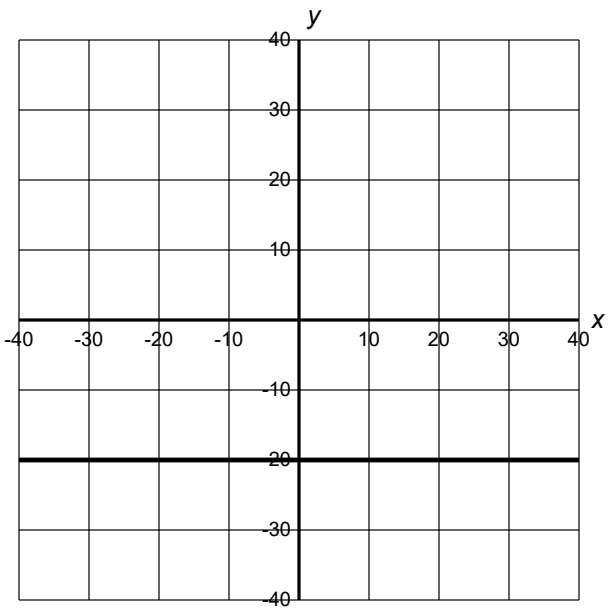
C. **$75 \le y \le 175$**

D. $0 \le x \le 7$

Range = the costs (y). 0 hours: $75. 5 hours: 75 + 20(5) = $175. Choices about x are the domain.

---

3. What are the equation and slope of the line shown on the grid?



F. $x = -20$; slope is zero

G. $y = -20$; slope is undefined

H. **$y = -20$; slope is zero**

J. $x = -20$; slope is undefined

Horizontal line: every y-value is −20, so y = −20. It never rises, so the slope is zero.

---

4. Marcus graphed $f(x)$ and $g(x) = f(x - 3)$ on the same coordinate grid. Which statement describes how the graphs of f and g are related?

A. The graph of the function $f(x)$ is translated up 3 units to produce $g(x)$.

B. The graph of the function $f(x)$ is translated down 3 units to produce $g(x)$.

C. The graph of the function $f(x)$ is translated right 3 units to produce $g(x)$.

D. The graph of the function $f(x)$ is translated left 3 units to produce $g(x)$.

A change inside the parentheses moves the graph sideways, opposite the sign: x − 3 → right 3.

---

5. What is the y-intercept of the linear function created by translating $f(x)$, represented in the table, 2 units right?

x	$f(x)$
−2	−7
−1	−4
0	−1
1	2
2	5
3	8

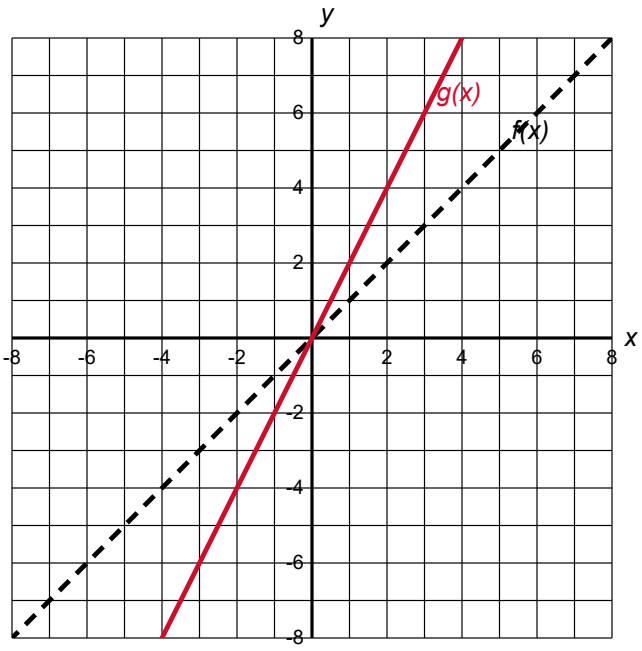
y-intercept = **−7**

Slide every x 2 right: the point (−2, −7) moves to (0, −7). Or g(x) = 3(x − 2) − 1 = 3x − 7.

---

6. Graph $g(x)$ based on the given transformation of $f(x)$.

$f(x) = x$ $g(x) = f(2x)$



g(x) = 2x: a line through (0, 0), (1, 2) and (3, 6)

f(2x) multiplies each input by 2, so the outputs double: slope 2.

---

7. What is the equation in slope-intercept form of the line that crosses the x-axis at −6 and is perpendicular to the line represented by $y = \frac{3}{2}x - 1$?

Equation: $y = -\frac{2}{3}x - 4$

Perpendicular slope = negative reciprocal of 3/2 = −2/3. Point (−6, 0): 0 = −2/3(−6) + b → b = −4.

---

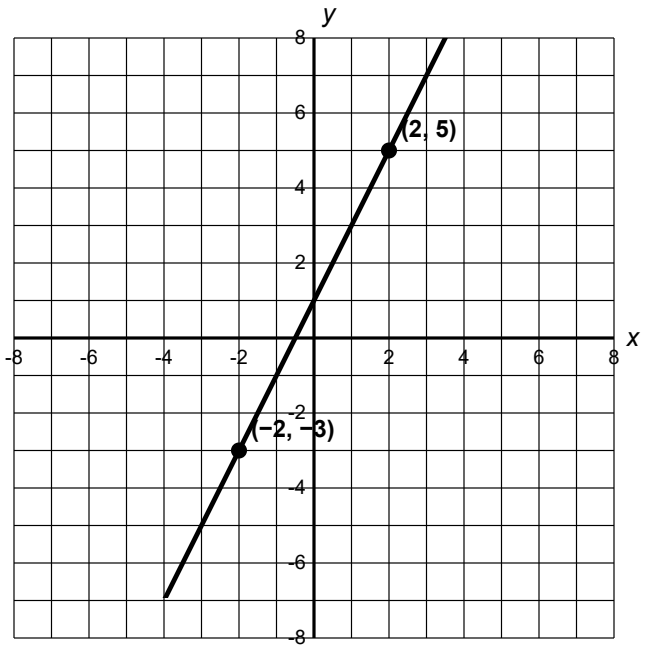
8. What is the equation in slope-intercept form of the line that passes through the point (5, 0) and is parallel to the line represented by $y = -2.4x + 1.6$?

Equation: ***y = −2.4x + 12***

Parallel → same slope −2.4. 0 = −2.4(5) + b → b = 12.

---

9. Write an equation in point-slope form that represents the graph.

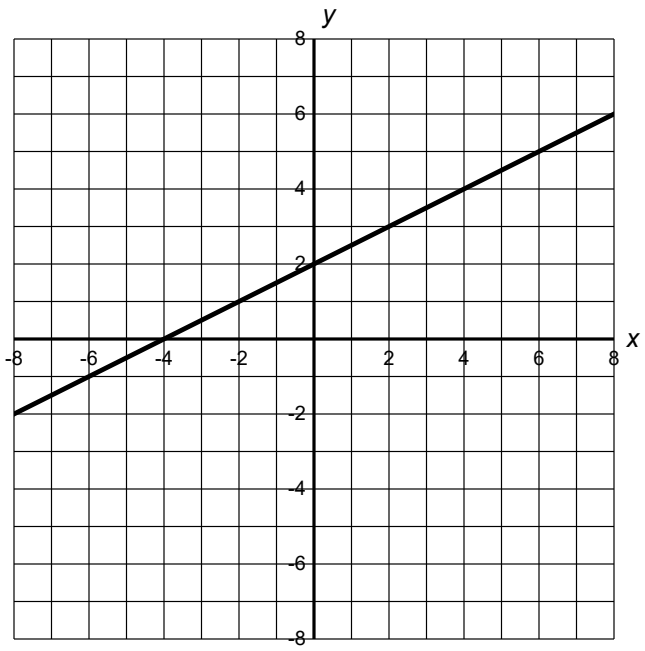


Equation: ***y − 5 = 2(x − 2)***

slope = (5 − (−3)) ÷ (2 − (−2)) = 8/4 = 2. Also correct: y + 3 = 2(x + 2).

---

10. The graph of a linear function is shown on the coordinate plane.



Which statements describe the graph of the linear function? **Select TWO correct answers.**

F. The graph is increasing.

G. The slope of the line is 2.

H. The *y*-intercept is (0, −4).

J. The zero of the function is −4.

K. The zero of the function is 2.

It rises left to right (increasing). It crosses the x-axis at −4, so the zero is −4. Slope = 2/4 = 1/2; y-intercept (0, 2).

Part 2 · Extra practice (Questions 11–14)

11. Let $f(x) = x - 1$, $g(x) = f(2x)$, and $h(x) = 2f(x)$. Which statement is true?

A. $g(x)$ and $h(x)$ are the same function.

B. **$g(x)$ and $h(x)$ have the same slope but different y-intercepts.**

C. $g(x)$ and $h(x)$ have different slopes but the same y-intercept.

D. $g(x)$ and $h(x)$ have different slopes and different y-intercepts.

g(x) = 2x − 1 and h(x) = 2(x − 1) = 2x − 2: both slope 2, y-intercepts −1 and −2.

---

12. Which pair of equations represents perpendicular lines?

F. $y = 5x + 1$ and $y = -5x + 2$

G. $y = -\frac{1}{3}x + 4$ and $y = 3x - 2$

H. $y = \frac{4}{5}x$ and $y = \frac{5}{4}x - 1$

J. $y = 2x + 3$ and $y = 2x - 3$

Perpendicular slopes are negative reciprocals: −1/3 × 3 = −1. (J is parallel.)

---

13. $f(x) = 2x + 1$ and $g(x) = f(x) - 4$. Write the equation of $g(x)$ and describe how the graph of f moves to make the graph of g.

$g(x) =$ **$2x - 3$**

Subtract 4 from every output: 2x + 1 − 4 = 2x − 3. The graph moves down 4 units.

Description: **translated down 4 units**

---

14. A tank holds 300 gallons of water. It drains at 15 gallons per minute until it is empty. Let x be the number of minutes. What is the domain of the function for this situation?

A. $0 \le x \le 300$

B. **$0 \le x \le 20$**

C. $0 \le y \le 300$

D. $15 \le x \le 300$

Domain = the minutes (x). Empty when 300 ÷ 15 = 20 minutes.
